

Modeling Freight Transportation as a System-of-Systems to Determine Adoption of Emerging Vehicle Technologies

A. Guerrero de la Peña¹; N. Davendralingam²; A. K. Raz³; V. Sujan⁴; D. DeLaurentis⁵; G. Shaver⁶; and N. Jain⁷

¹School of Mechanical Engineering, Purdue Univ., West Lafayette, IN 47907-2099. E-mail: guerrer2@purdue.edu

²School of Aeronautics and Astronautics, Purdue Univ., West Lafayette, IN 47907-2099

³School of Aeronautics and Astronautics, Purdue Univ., West Lafayette, IN 47907-2099

⁴Cummins, Inc., Columbus, IN 47201

⁵School of Aeronautics and Astronautics, Purdue Univ., West Lafayette, IN 47907-2099

⁶School of Mechanical Engineering, Purdue Univ., West Lafayette, IN 47907-2099

⁷School of Mechanical Engineering, Purdue Univ., West Lafayette, IN 47907-2099

ABSTRACT

The U.S. freight transportation system is a complex agglomeration of interacting systems that includes line-haul and urban delivery vehicles, inter and intra-city highways, and support infrastructure. In order to project the evolution of the system and the market penetration of emerging freight vehicle technologies, it is important to model the aforementioned interconnections, public adoption preferences, and operational and policy constraints that impact it. In this paper, we propose a system-of-systems engineering approach to define the scope of influential mechanisms and abstract an appropriate model of the U.S. freight transportation system with focus on a line-haul scenario. Implementation over a multi-city network is posed as a constrained mixed-integer linear program. The allocation and operation of three vehicle architectures—conventional diesel, diesel platooning, and battery electric—are optimized over a multi-city network to minimize the fleet-wide total cost of ownership over a twenty-year time horizon. We examine the effects of projected changes in energy cost, freight demand, and hours-of-service regulations on the annual market share evolution of these technologies.

INTRODUCTION

Motivation and problem definition: More than 90% of Heavy Duty Class 8 vehicles used for line-haul operation are still dieselized and are a significant producer of carbon emissions in the U.S (National Research Council, 2010) (U.S. Department of Transportation, 2016). Over the last few decades, stringent regulations and improved vehicle technologies have been introduced to reduce national fuel consumption and CO₂ emissions. Technologies that increase the aerodynamic efficiency of the vehicles, reduce rolling resistance, or increase engine efficiency have already been widely adopted (North American Council for Freight Efficiency, 2016). Through these technologies, vehicle fuel efficiency has seen an average increase of 12% in the U.S. (NACFE, 2016). However, these technologies only offer incremental improvements in the fuel economy of conventional Diesel vehicles. New and revolutionary architectures, including alternative fuel vehicles, hybridization, full electrification, and increased levels of autonomy have been proposed to further decrease national fuel consumption and vehicle emissions (Vimmerstedt et al., 2015). A tool that can project the adoption paths of these emerging technologies would enable technology manufacturers and policy makers to make decisions regarding technology innovation, introduction to market, and economic incentives that will

maximize adoption and result in targeted reductions in fuel consumption and CO₂ levels.

Gaps in literature: The cost attractiveness of emerging freight transportation technologies and the ensuing reduction in emissions given future adoption is an issue that has been addressed in several studies. Lammert et al. (2014) present fuel consumption reduction of two Class 8 Diesel tractor-trailer vehicles operating in platooning mode at varying gap distances, steady-state speeds, and gross vehicle weight during a series of track tests. Their results demonstrate combined fuel savings of up to 6.4%, showing an attractive return on investment with respect to operational fuel costs. Zhao et al. (2013) evaluate various powertrain architectures, including Diesel conventional, hybrid, electric, and natural gas, over day-long, short-haul, and long-haul simulated drive cycles. The authors derive the CO₂ emissions reduction potentials given adoption of these technologies and present the break-even fuel costs for economic attractiveness. Fulton and Miller (2015) explore deep market penetration scenarios of different low-carbon vehicle technologies, such as alternative fuel architectures and electric vehicles, and the resulting capability to meet 80% reduction of CO₂ emissions in the U.S. by 2050. Their results are based on historical and projected data for freight demand, trucking tonnage, and mile share in the U.S. Finally, Schafer and Jacoby (2006) present rates of adoption for different personal vehicles and Heavy Duty Class 8 Diesel truck technologies under CO₂ emissions constraints and penalty costs. Despite introducing interconnections between economic factors, CO₂ emissions policies, and vehicle performance to their analysis, Schafer and Jacoby extrapolate trucking metrics, such as average annual miles and consequently truck fuel consumption, from historical data. Effects on the evolution of freight transportation metrics caused by policy changes and introduction of new technologies may not be effectively captured by projection of historical trends.

Studies like (Lammert et al., 2014) and (Zhao et al., 2013) determine economic attractiveness based only on vehicle efficiencies over an isolated drive cycle without considering external factors that will also affect technology adoption. Fulton and Miller (2015) project national reduction in CO₂ emissions provided assumed technology adoption scenarios, while Schafer and Jacoby (2006) do so by extrapolating historical trucking data. These studies do not provide a holistic treatment of how fleets adopt and operate vehicles with different technologies over the transportation network. The coupling of vehicle efficiencies and freight transportation system considerations makes technology adoption a multi-faceted problem. These interconnections must be simulated to effectively project the operational costs, technology adoption paths, and resulting emissions outcomes. *Achieving this goal requires a simulation framework with an ability to include factors pertaining to multiple independent systems—factors that cannot be observed in a single drive cycle—such as fleet management and operation, policies, and network characteristics that ultimately affect economic attractiveness of emerging technologies.*

Contribution: In this paper, we propose a formulation to estimate fleet-wide operational and purchasing costs as a function of vehicle technology selection and allocation over a regional freight transportation network in order to project future adoption of emerging technologies. A System-of-Systems (SoS) engineering methodology is used to define the scope of considerations necessary to model adoption of freight transportation technologies, as well as the appropriate level of abstraction for simulation. Reducing the complexity of vehicle performance modeling enables introduction of fleet management factors, policies, and infrastructure availability considerations that influence the attractiveness of fleet transportation technologies. Moreover, this approach produces a holistic analysis of technology adoption trends, in contrast to traditional methods of platform-centric analysis.

Outline: This paper is organized as follows. The next section includes a brief overview of

the System of Systems (SoS) methodology applied to a regional line-haul freight transportation system. The linear program implemented for optimization of operational and purchasing costs of a regional fleet is described in the Model Formulation section. Technology adoption projections for a single fleet operating over a small network of regional highways are presented in the Simulation section, followed by concluding statements.

SYSTEM OF SYSTEMS FORMULATION

A System of Systems is a complex system that consists of a collection of entities which collaborate for a unique purpose while retaining operational and managerial independence. An SoS will also show evolutionary dynamics, that is, their internal structure will change over time as the constituent systems and networks evolve. The U.S. line-haul freight transportation system shows a prevalence of these traits, it is composed of interconnected systems of vehicles, inter and intra-city highways, and support infrastructure organized at multiple levels and evolving over time. In order to model the evolution of this system and determine adoption of emerging freight vehicle technologies, it is important to identify relevant holistic factors and mechanisms and translate them systematically into an actionable model. Recognizing the need for a holistic methodology, DeLaurentis (2005) has developed a System-of-Systems modeling and analysis framework that can be used to view the transportation system within a SoS context. This SoS framework has three main phases—the *definition* phase, the *abstraction* phase, and the *implementation* phase—as discussed below.

Definition Phase: The definition phase seeks to establish a structural map of the SoS—in this case, the regional freight transportation system—in terms of hierarchies and categories. This phase uses a construct known as a ROPE table, shown in Table 1, which serves as a problem scoping platform for subsequent analysis. The columns of the ROPE table correspond to the Resources, Operations, Policy and Economics dimensions of the SoS space. Each row represents the hierarchical levels of the SoS. In this case, the alpha level of the ROPE table corresponds to the discrete technologies that are implemented in line haul vehicles. Higher levels, beta and above, reflect aggregations of elements from lower level entities; for example, a combination of technologies at the alpha level would constitute a vehicle unit at the beta level and a combination of beta-level vehicles would constitute a fleet at the gamma-level. The ROPE table enables the SoS engineer/designer to seek out various factors across ROPE categories that are expected to play a role in the projected evolution of the SoS and its future composition of vehicle technologies. The entries shown in Table 1 are representative of a U.S. regional Line-Haul system. For example, when building the ROPE table for line-haul transportation, driver hours of service limits, weight, speed, and emission restrictions are identified as influential U.S. policy considerations at different levels of the SoS. This implies that evaluation of powertrain or vehicle technology performance alone, detached from the rest of the SoS levels, is not enough to observe evolution of the system.

Abstraction Phase: The abstraction phase includes descriptors for the entries of the ROPE table and considerations for stakeholder behaviors, incentive structures, and relevant models used to describe them. In the context of our line-haul problem, the total cost of ownership is a utility that fleet owners seek to optimize when considering vehicle purchase. An examination of the ROPE table reveals that the total cost of ownership (listed under Economics at the gamma-level) is influenced by more factors than the cost of emerging vehicle technologies alone. The goal of the abstraction phase, then, is to develop representations that will describe the total cost of ownership and incorporate the influence of such relationships so an appropriate strategy for

implementation can be brought to bear in the next phase. This problem abstraction is provided in the form of a ‘paper model’, i.e. a description of the big picture dynamics, as shown in Figure 1. The paper model facilitates the transition to the implementation phase by defining the organization and interconnections of the overall entities described in the ROPE matrix.

Implementation Phase: The implementation phase seeks to realize solution approaches for the problem formulation obtained in the abstraction phase. In this work, we scope the profit-seeking behavior of a representative line-haul fleet in the abstraction phase. We then model this behavior in the context of a cost minimization optimization problem and write the relevant mathematical expressions based on the variables and descriptors established in the abstraction phase. The mathematical formulation and implementation are described in the following section.

Table 1. Regional Line-Haul Freight Transportation System ROPE matrix

	Resources	Operations	Economics	Policy
Alpha	• Diesel engine • Battery Electric	• Powertrain fuel consumption	• Cost of fuel • Cost of energy	• Emission restrictions
Beta	• ICE Conventional vehicle • ICE + Platooning • BEV	• Ton-mi/gal efficiency • Average day operation • Vehicle life cycle • Cargo load/capacity • Miles driven based on selected routes • Operate at constant speed over route • Operator hours • 2-veh platooning • Vehicle range	• Cost of fuel, energy consumed • Cost of purchase • Cost of driver/hour	• 80,000 lb. weight limit
Gamma	• Vehicles in single regional fleet	• Fleet distribution • Fleet size • Vehicle replacement cycles and years of service limits	• Total cost of ownership decision metrics per year	• Driver hours of service limits
Delta	• 4-city network: • City 1-City 2- • City 3- City 4	• Total Freight demand between cities by weight • Traffic conditions: vehicles on road, road density capacity, travel time • Single direct route between cities	• Cost of fuel • Cost of energy	• Speed limit • Regional emissions reduction

MODEL FORMULATION

This study focuses on line-haul truck operation over a regional inter-city network as defined in the ROPE matrix and abstraction model (Table 1 and Figure 1, respectively). The computer model can be used to simulate the decision process of fleets to purchase those vehicle architectures that are economically attractive to them given their operational and purchase costs. Our intent is to replicate the Total Cost of Ownership (TCO) minimization behaviors of fleet

owners for Heavy Duty Class 8 trucking highway operation given vehicle highway performance, fleet management considerations, infrastructure availability, and external influences such as cost of energy and regional freight demand. A system optimal traffic allocation approach is used to estimate the operational costs, while new vehicle purchase costs and turnover sales revenue are included as metrics for vehicle acquisition. Fleet management, policies, and vehicle operational considerations are formulated as constraints. The linearity of the resulting model's objective function and constraints makes the optimization problem a Mixed-Integer Linear Program (MILP), to which highly efficient and matured means of solution are available.

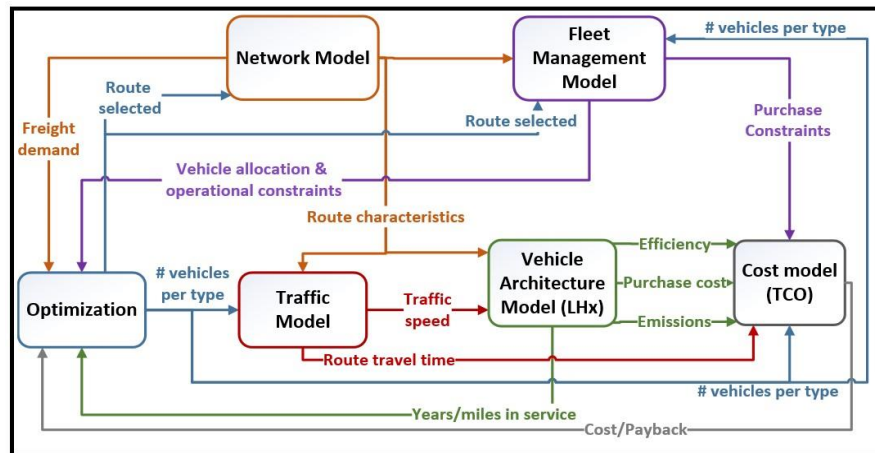


Figure 1. Regional Line-Haul Freight Transportation System Paper Model.

Problem Formulation: A mixed-integer linear programming (MILP) formulation is chosen to represent a large regional fleet with a focus on the line-haul highway operation considerations identified in the ROPE matrix. The objective function represents vehicle purchasing criteria as a function of estimated economic attractiveness of different technologies operated over the network routes. The decision variables are defined as x_{qn}^s , representing the number of n vehicles of type q that originate at s per year, $x_{qn,ij}^s$, vehicle n of type q originating at s traveling on link (i,j) , and y_{ij}^h , the cargo link flow originating at h .

Objective Function: The objective function represents the Total Cost of Ownership (TCO) criteria commonly used by fleet owners in order to select vehicles for purchase. On average, fuel consumption, repair and maintenance of a vehicle, and driver wages incur the highest percentage of total operational costs on a per km basis over a vehicle's lifecycle (Torrey and Murray, 2016)(NACFE, 2016)(Transportation Research Board, 2011). Fleet owners compute these cost components as a decision-making criteria for purchase of new technologies over conventional ones. The TCO objective is comprised of these operational costs, the costs associated with vehicle technology reliability, and the cost of purchase and turnover sales revenue. Fleets commonly purchase vehicles on a yearly basis, and therefore this decision-making process is exercised annually throughout the period of study. The objective function is defined as the total fleet operational and purchasing costs, such that: $J_k = C_{op} + C_{pur}$. Here, the subscript k indicates that the total cost of ownership is computed every year, where $k = [1, 20]$ for the case study presented. To simplify the notation, the subscript k is dropped throughout the formulation with the exception of cases where required to indicate the use of values from previous years. The cost

of operation includes the cost of energy consumed C_{eq} , drivers C_{dr} , maintenance C_M , and revenue losses C_R , such that $C_{op} = C_{ec} + C_{dr} + C_M + C_R$.

Table 2. Model Parametrization

Beta Level-Vehicle Architecture Parametrization		
Operational		
ξ_q	Efficiency function of vehicle type q	$\frac{gal}{km}$
R_q	Driving range of vehicle type q	km
W_q	Capacity for vehicle type q units of weight	ton
$E_{q,co2}$	CO_2 emission rates for vehicle architecture q	$\frac{g}{km}$
B_q	Reliability of vehicle architecture q	$\frac{\%trips}{year}$
Economic		
$C_{M,q}$	Cost of maintenance per mile	$\frac{\$}{km}$
$C_{p,q}$	Cost of purchase for vehicle type q	$\$$
$C_{r,q}$	Resale value for vehicle type q	$\$$
Gamma Level-Fleet Management Parametrization		
l_{min}, l_{max}	Vehicle turnover range	years
γ	Projected vehicle life-cycle period	years
C_{driver}	Driver wages	$\frac{\$}{km}$
B_f	Fleet's budget for vehicle purchase	$\$$
$C_{delay,c}$	Proportion of revenue loss due to delay for cargo type c	$\frac{\$}{hr}$
Delta Level-Network Parametrization		
b_i^h	Cargo demand from origin h to destination i	ton
d_{ij}	Length (distance) of link (i,j)	km
h_{os}	Hours of service limit	hours
C_{eq}	Cost for energy consumed by vehicle type q	$\$$

Energy consumption costs vary depending on the vehicle technology used. Cost of energy is defined as $C_{ec} = \gamma \sum_q \sum_{(i,j) \in A} x_{q,ij} d_{ij} C_{q,ij}$, where $x_{q,ij} = \sum_s \sum_n x_{qn,ij}^s$ represents the flow of vehicles of type q over highway link (i,j) regardless of their origin, and A is the set of city-nodes in the

network. The cost of energy consumed per mile, $C_{q,ij} = \xi_q(u_{ij})C_{eq}$, is a function of the vehicle efficiency, ξ_q , which depends on average vehicle speed. Operational costs are computed as a function of the estimated total number of trips in an average operational day. In order to estimate lifecycle costs, the cost of energy consumed over an average day is multiplied by γ , the number of years in a vehicle's lifecycle period.

The driver costs, C_{dr} , are computed on a per km basis, given the total distance traveled by fleet vehicles on an average operational day. Similar to the energy consumption costs, the driver costs are weighted over the vehicle's expected lifecycle: $C_{dr} = \gamma \sum_q \sum_{(i,j) \in A} x_{q,ij} d_{ij} C_{driver}$. The

technology type and age of a vehicle can affect its maintenance and repair costs and is commonly used as a metric by fleet owners to identify the appropriate turnover age of their vehicles.

Maintenance costs are defined as $C_M = \gamma \sum_q \sum_{(i,j) \in A} x_{q,ij} d_{ij} C_{M,q}$. Reliability of cargo delivery may be

affected by vehicle or component break-down. Here, we assume technology reliability issues result in time delays, $T_{d,q}$, and scheduling of a second vehicle for completion of delivery is not necessary. In that manner, reliability costs are modeled as losses in revenue due to the incurred delay and are a function of both vehicle and cargo type: $C_R = \gamma \sum_q \sum_{(i,j) \in A} x_{q,ij} B_q T_{d,q} C_{delay,c}$.

The cost of purchase considers the cost of buying new vehicles and the revenue generated by selling used ones: $C_{purch} = C_{nv} - C_{sr}$. Fleets will purchase new vehicles 1) to replace those beyond their economic lifecycle or 2) to increase fleet volumes due to an increase in freight demand. Here we assume that all vehicles are purchased new, such that $C_{nv} = \sum_q x_{q,new} C_{p,q}$. The

variable $x_{q,new}$ is introduced to represent the vehicles of technology type q newly adopted in the current year of projection, k . This means there are additional vehicles of type q originating at node s that were not allocated in previous years. The variable $x_{q,new}$ is given by $[x_q]_k - [x_q]_{k-1}$, where $x_q = \sum_s \sum_n x_{qn}^s$. Furthermore, $x_{q,new}$ is positive only if new vehicles are allocated to origin s ,

and zero otherwise. Fleets will sell older vehicles when they are near the end of their economic life, the age at which maintenance and repair costs increase and efficiency is no longer optimal. At this point, fleets may replace older vehicles with a newer purchase. The revenue obtained from a sale is computed as $C_{sr} = \sum_q x_{q,r} c_{r,q}$ and then implemented as an offset to the purchasing

budget of the current year.

The turnover period, $[l_{min}, l_{max}]$, during which a vehicle approaches the end of its economic life and is considered for replacement, varies by fleet. Here we assume that a line-haul fleet has a fixed range for vehicle turnover age. In contrast to new vehicles purchased, the variable $x_{q,r}$ is given by $[x_q]_{k-1} - [x_q]_k$ and is introduced to represent vehicles sold by the fleet. The value is positive if vehicles of type q allocated to origin s during the current year of projection are less than in the previous year, and zero otherwise. Computation of $x_{q,r}$ must take precedence over new vehicle purchase, as older vehicles may be sold and replaced with newer vehicles of the

same technology. In summary, the TCO is computed as follows:

$$J = \gamma \sum_q \sum_{(i,j) \in A} x_{q,ij} \left[d_{ij} (C_{q,ij} + C_{driver} + C_{M,q}) + B_q T_{d,q} C_{delay,c} \right] + \sum_q x_{q,new} C_{p,q} - \sum_q x_{q,r} C_{r,q} \quad (1)$$

Constraints: The vehicle demand over the network is defined as a function of cargo demand, b_i^h , between (h,i) city pairs. Vehicle link flow will be optimized in order to satisfy cargo demand, vehicle flow balance entering and leaving nodes, and capacity constraints as given by Equations (2a)-(2d). Hours of service limit, h_{os} , as shown in Eq. (2e), will also have an effect on the amount of vehicle trips taken within the time constraint, and therefore the number of vehicles needed to be allocated over the network. An intermediate binary variable, x_{qn} , is introduced and assigned a value of 1 if the n th vehicle of type q is used. This assists in the computation of total number of vehicles of type q purchased and allocated to city s , such that $x_{qn}^s \leq M \sum_j x_{qn,ij}^s$ and

$$Mx_{qn}^s \geq \sum_j x_{qn,ij} \text{ for all } i = s, \text{ where } M \text{ is a sufficiently large number.}$$

Table 3. Optimization Constraints

Constraint	Expression	Constraint	Expression
(2a)	$\sum_j y_{ji}^h - \sum_j y_{ij}^h = b_i^h$	(2d)	$x_{qn,ii}^s = 0, \quad y_{ii}^h = 0$
(2b)	$\sum_h y_{ij}^h \leq \sum_s \sum_q \sum_n x_{qn,ij}^s W_q$	(2e)	$\sum_i \sum_j x_{qn,ij}^s t_{r,ij} \leq h_{os}, \quad \forall s, q, n$
(2c)	$\sum_j x_{qn,ji}^s - \sum_j x_{qn,ij}^s \geq 0, \quad i \neq s$		

New vehicle purchases are constrained by a user-defined fleet budget, which is offset by the revenue created from vehicles sales, such that $\sum_s \sum_q x_{q,new}^s C_{p,q} - \sum_s \sum_q x_{q,r}^s C_{r,q} \leq B_f$. A market

penetration constraint, $\sum_s x_{q,new}^s \leq Q_{available} \quad \forall q \in Q$, is also added to represent the availability of

vehicle technologies entering the market. The parameter $Q_{available}$ can be calibrated to limit the rate of penetration of newer technologies with lower production rates as existing technologies are phased out. Vehicle resale is also constrained such that

$x_{q,new}^s (t_{yk} - l_{max}) \leq x_{q,r}^s \leq x_{q,new}^s (t_{yk} - l_{max}) + x_{q,new}^s (t_{yk} - l_{min})$. Vehicles older than the maximum allowable age will be sold, while vehicles within the turnover range may be considered for replacement.

Some technologies may have a significant impact on the operation of vehicles over the network. For two-vehicle platooning operation, vehicles with this technology must travel in pairs over any link (i,j) in order to gain the associated efficiency benefits, regardless of their origin or destination. Therefore, an intermediate integer variable, P_{ij} , is introduced to enforce this constraint, such that $\sum_s \sum_n x_{qn,ij}^s = 2P_{ij}$ for any architecture q with platooning capability.

We implement the method and equations introduced by Zheng et al. (2017) to determine feasibility of travel, e_{ij}^s , for electric vehicles originating at s traveling on link (i,j) , given the vehicle range and location of charging stations. As summarized in (Zheng et al., 2017), the

variable L_j^s is incremented by the (i,j) link's distance d_{ij} if node i does not have charging infrastructure. The variable e_{ij}^s will then have a value of 1 if the total distance L_i^s is within the vehicle's range limits. The location of charging stations is not optimized in our formulation; it is instead defined as a network parameter. Therefore, the status, E_i , of a node as a charging station is an input to the MILP.

Table 4. Electric Vehicle Constraints

Constra int	Expression	Constrai nt	Expression
(3a)	$L_j^s \geq L_i^s + d_{ij} - M(1 - e_{ij})$	(3d)	$x_{qn,ij}^s \leq M e_{ij}, \quad \forall n$
(3b)	$L_i^s \leq R_q$	(3e)	$L_i^s \geq 0, \quad L_i^{s'} \geq 0$
(3c)	$L_i^{s'} \geq L_i^s - M E_i, \quad L_i^{s'} \leq L_i^s + M E_i,$ $L_i^{s'} \leq M(1 - E_i)$	(3f)	$e_{ij} \in \{0,1\}, \quad E_i \in \{0,1\},$ $E_i = \begin{cases} 1 & i \in C \\ 0 & \end{cases}$

Traffic Model: Vehicle efficiency will vary with respect to average vehicle speed over a network route. The solution u_{ij} to Greenshield's macroscopic traffic flow model (Williams, n.d.) provides the average traffic speed based on the number of vehicles, $q_{f,ij}$, both freight and passenger, introduced to the link and the route characteristics such that:

$-\frac{k_{ij}}{v_{f,ij}} u_{ij}^2 + k_{ij} u_{ij} - \frac{q_{f,ij}}{N_{ij}} = 0$. The Greenshield equation involves a nonlinear term with respect to the traffic speed, u_{ij} , itself a function of freight traffic flow. In order to have a linear constraint to facilitate the use of a linear solver, we estimate the number of freight vehicles traveling over each link (i,j) and solve for the speed a-priori. Finally, route time can be computed as $t_{r,ij} = \frac{d_{ij}}{u_{ij}}$.

Emissions Model: All vehicle architectures selected as part of this study must be compliant with the CO₂ emissions and fuel standards as specified by the EPA Greenhouse Gas (GHG) Phase 2 release (US EPA, 2016). However, computation of cumulative regional emissions can be useful to determine policy sustainability and future carbon reduction given the displacement of fossil fuel emissions by greener technologies. It can also assist in the determination of necessary cost incentives for customers in order to achieve a desirable outcome. The regional emissions, E_R , are calculated as a cumulative regional value, given by $E_R = \sum_q \sum_{i,j \in A} x_{q,ij} d_{ij} E_{q,co2}$. The vehicle

emissions output, $E_{q,co2}$ in $\left[\frac{g}{km} \right]$, provides the mass of CO₂ produced by the energy consumed

per kilometer. In the case of Diesel engines, this is proportional to the gallons of fuel used, whereas electric vehicle emissions are a function of the mass of CO₂ produced per kWh

consumed: $E_{q,co2} = \frac{CO_2}{kWh} \xi_q$. The value of $\frac{CO_2}{kWh}$ can vary with respect to the year of projection to represent a regional shift to cleaner sources for production of energy.

MODEL SIMULATION

A 4-city network is defined to demonstrate the capability of the proposed model; the cargo demand and route distances are shown in Table 6. Three vehicle technologies—Diesel, Diesel platooning, and battery electric—are evaluated with the parametrization shown in Table 5. Vehicle capacity and maintenance costs are constant across all technologies to reduce complexity of model calibration and analysis. To limit computation time, only cities 1 and 2 are chosen as origins for vehicle allocation and locations of charging stations. This and the assumed electric vehicle range limit electric vehicle operation to the direct routes between cities 1 and 2. The cost of electricity is maintained from year to year at \$0.10 per kWh, while the cost of Diesel is assumed to vary. The analysis shows the purchasing behavior of a small regional fleet over a 20 year period assuming constant cargo demand, a vehicle turnover range of 3-5 years, and vehicle lifecycle, γ , of 5 years.

Figures 2 and 3 show the effects of the hours of service (HOS) policy, fleet purchasing budget constraints, and cost of Diesel on vehicle adoption projections, vehicle miles traveled, and resulting CO₂ emissions for a single fleet. All adoption, VMT, and emissions values are normalized with respect to the Diesel vehicle outputs for the year 2017. Although long-term Diesel prices are difficult to estimate, the selected values for simulation, shown by the dashed curve in Figure 2, serve to demonstrate the effect of fuel cost volatility on technology selection.

Table 5. Vehicle Architecture Parametrization

Veh. Type	Peak Eff. (mi/EC)	W_q (ton)	$C_{p,q}$ (\$k)	Depreciation Rate (%/year)	R_q (mi)	Emissions kg CO ₂ /EC	B_q (% trips delayed)
Diesel	6.5	20	160	0.1	1000	10.34	1%
Diesel	6.8	20	172	0.2	1045	10.34	4%
Platooning							
BEV	0.34	20	400	0.2	240	0.4	5%

Table 6. Network Parametrization

Cargo Demand (ton/day)					Route distance (mi)				
O/D	City 1	City 2	City 3	City 4	O/D	City 1	City 2	City 3	City 4
City 1	0	120	100	0	City 1	0	182	297	470
City 2	100	0	80	80	City 2	182	0	242	289
City 3	0	0	0	0	City 3	297	242	0	309
City 4	0	0	0	0	City 4	470	289	309	0

We can observe in all cases shown in Figure 2 that the adoption of platooning vehicles increases in the year 2022 as the cost of Diesel increases. However, technology adoption does not vary between 2022 and 2031 regardless of variation in cost of Diesel. This indicates that *fleet adoption of more efficient technologies may be delayed following a large but temporary change in the cost of Diesel due to vehicle turnover restrictions*. On the other hand, an increase in

adoption of conventional Diesel vehicles occurs from 2031-2036 as the price of Diesel remains low.

An increase in hours of service, from 11 to 15 hours, in addition to a reduction in fuel costs, increases the adoption of conventional Diesel vehicles toward the end of the simulation period, as indicated by a comparison between Figures 2a and 2b. However, no effects are observed on vehicle adoption in the first 15 years.

As expected, *a limited purchasing budget will impact the adoption of platooning vehicles* as indicated by comparison of Figures 2a and 2c, and 2b and 2d. Again, Figures 2b and 2c show the same adoption trends despite changes in budget and hours of service, yet different VMT and emissions values. Upon further inspection of Figures 3a and 3b, we observe that *vehicle utilization varies with the increase in hours of service*, as indicated by the vehicle miles traveled (VMT) per year. Adoption values are equivalent in the first 15 years (for a fixed number of hours of service (HOS)), but resulting emissions differ between the two cases from 2017 to 2021 (compare Figures 3a and 3c). Note that emissions are plotted on the right y-axis in Figure 3. As shown in Figures 2d and 3d, *a change in budget and HOS restrictions causes the most variation in technology adoption* throughout the 20 year period.

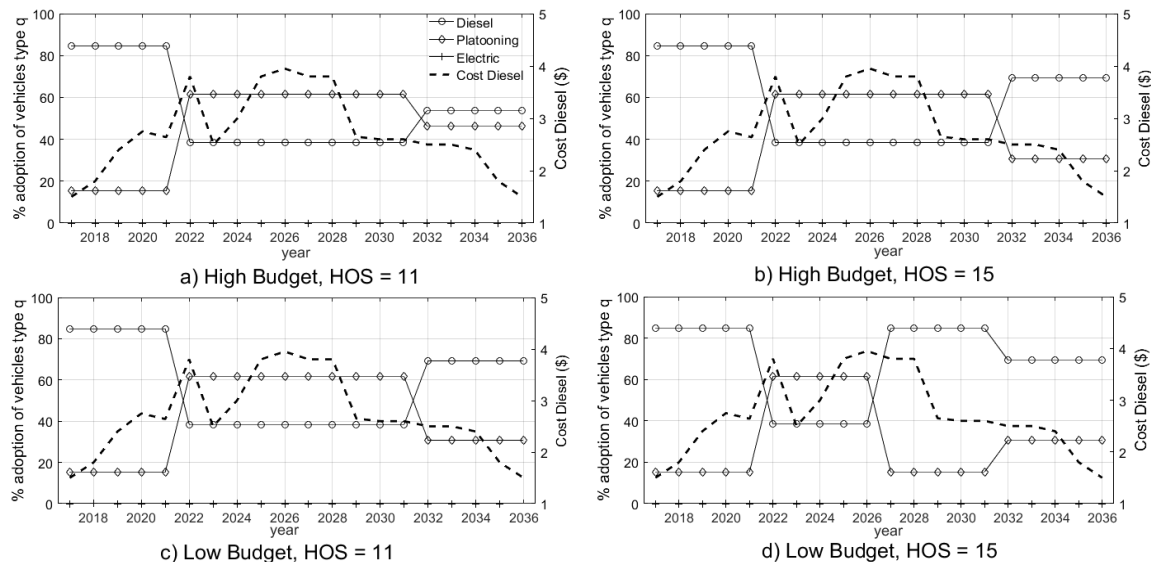


Figure 2. Annual vehicle adoption projections for different budget and HOS values. Cost of Diesel is plotted using a dashed line. Adoption values are normalized.

Figures 2 and 3 show electric vehicles are not adopted in the simulated scenarios due to the high purchase cost. Figures 4 and 5 show the impact of decreasing the purchase cost of electric vehicles from \$400,000 to \$300,000 on adoption and fleet emissions, as well as the effects of hours of service regulations. For this case study, we assume a regional dependency on coal and gas as sources of electricity and estimate a production rate of 0.4kg of CO₂ per kWh consumed by battery electric vehicles. We observe that following a decrease in cost, electric vehicles are adopted between the years 2022-2031, as shown in Figures 4a and 4b, but adoption drops to zero during the last 5 years of study when the cost of Diesel has remained low. Figure 4b shows that *an increase in HOS, from 11 to 15 hours, results in increased adoption of electric vehicles* from 2022-2031. As a result, CO₂ emissions are lower throughout this 10 year period, as observed in Figure 5b.

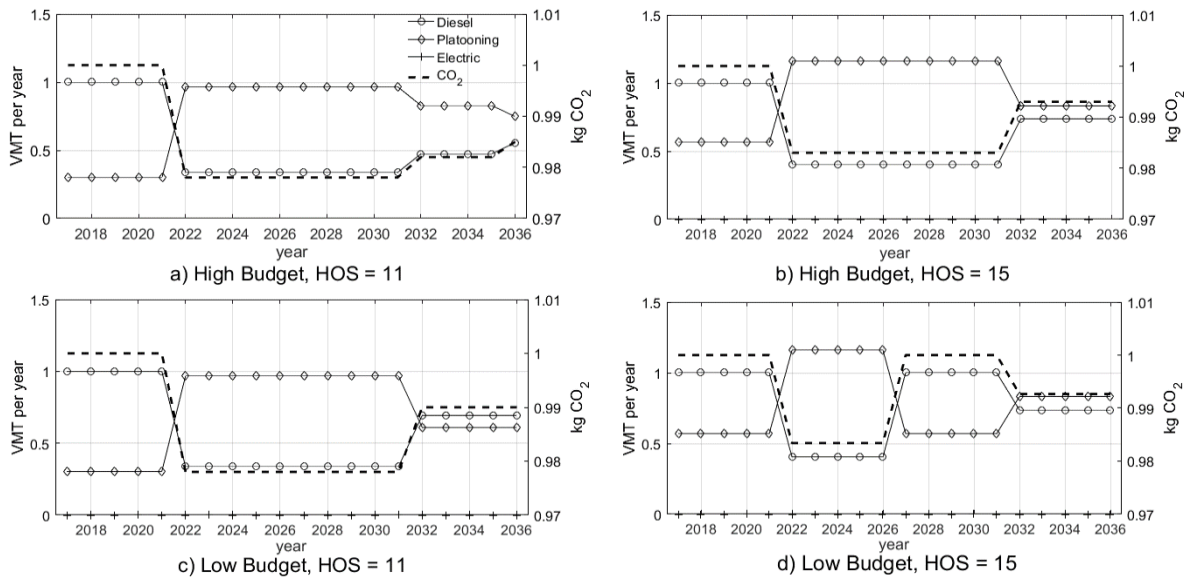


Figure 3. Annual total VMT projections per vehicle type for different budget and HOS values. Values are normalized.

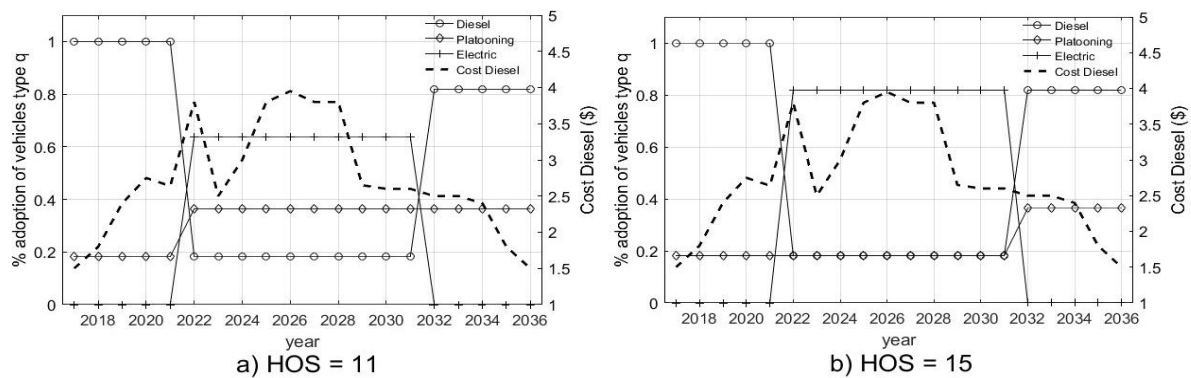


Figure 4. Annual vehicle adoption projections for different HOS values and reduced BEV purchase cost. Adoption values are normalized.

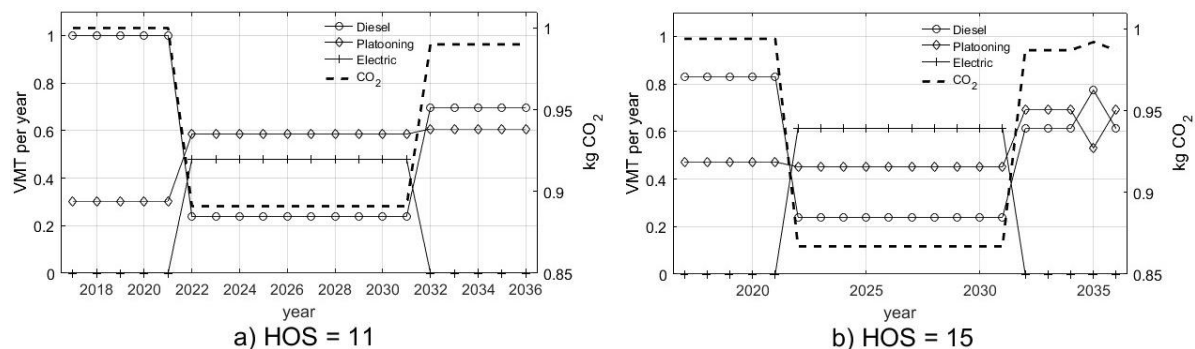


Figure 5. Annual total VMT projections per vehicle type for different HOS values and reduced BEV purchase cost. Values shown are normalized.

CONCLUSION

U.S. freight transportation is a complex system-of-systems; it is composed of interconnected systems including line-haul and urban delivery vehicles, inter and intra-city highways, and

support infrastructure. In this paper, we used the SoS engineering methodology to define, abstract, and simulate the U.S. freight transportation system with a focus on line-haul scenarios. We proposed a constrained mixed-integer linear program to optimize the allocation of three vehicle technologies—conventional Diesel, Diesel platooning, and battery electric—over a multi-city network with respect to minimization of total cost of ownership over a twenty-year time horizon. The effects of energy cost, freight demand, and hours-of-service regulations were evaluated to determine annual market share evolution of these technologies. The results demonstrated the sensitivity of future adoption trends to changes in exogenous factors identified during the SoS definition phase – fuel costs, fleet budget and vehicle turnover considerations, and hours of service policies. Future work can focus on increasing the fidelity of implementation and extend the formulation to simulate vehicle allocation of several fleets over a larger inter-city region. This would enable the user to observe regional adoption trends of emerging technologies provided a range of fleet management considerations and representative distribution networks.

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